Lexical versus Semantic Skill-Gap Measurement: A Sensitivity Audit on Job-Posting Data

Author Names Redacted for Double-Blind Peer Review
Department of Computer Science and Engineering
Academic Research Institution
contact@university.edu

***Abstract*—Reports that computing curricula and employer demands diverge are common, yet measuring either at scale remains limited. We compare a 1.35-million-posting LinkedIn corpus—a single scrape family—against an employer-derived occupational reference profile (1,167 postings, five occupational categories). Analyses use a 519,814-posting nested subset; both indices are re-checked on a 1.29-million-posting superset, and a title-free join finds no profile/market pairs at Jaccard ≥ 0.7. N-gram extraction over an eight-category taxonomy combines Sentence-BERT alignment at four cosine thresholds ($\tau \in$ 0.4, 0.5, 0.6, 0.7; 1,000-fold bootstrap). Exact string matching yields a Skill Gap Index (SGI) of 52.0%; semantic alignment reduces this by 20 percentage points to 32.0% at $\tau = 0.5$ (95% CI [23.0%, 41.0%]). Both are the headline configuration ($K = P = 100$); on the primary corpus, sweeping (K, P) keeps lexical SGI in [0.52, 0.73] and semantic SGI at $\tau = 0.5$ in [0.26, 0.42]. The profile covers 30.4% of the CS2023 Body of Knowledge by embedding coverage—a computational anchor, not a human audit—so no curricular gap size is claimed. Reference-profile deficits concentrate in Cloud Infrastructure (2.45 pp lexical, 0.94 pp semantic) and Programming & Development (1.69 pp lexical); soft-skill requirements are fully covered. The matcher agrees with expert ESCO mappings at $\kappa = 0.737$ (precision 91.9%, recall 75.7%), though the gold set skews to software development; the lexical-to-semantic reduction is consistent with vocabulary divergence accounting for part of the gap. A taxonomy-derived ONET×BLS profile is vocabulary- and scope-dominated, not comparable to postings at skill or category level.**

***Index Terms*—Skill-gap measurement, Job-posting analytics, Employability, Natural Language Processing, Sentence-BERT, Measurement validity, Data provenance.**

I. Introduction

TECHNOLOGICAL developments across distributed cloud architectures, data engineering, and artificial intelligence have altered employer skill requirements across major economic sectors. Because university curricula are updated through multi-year institutional review cycles, curricula and market skill demands plausibly change at different rates—a difference this paper measures rather than presumes [1], [2]. Educational researchers and labor economists label this *perceived* difference the *skills gap*—defined as the divergence between competencies taught in degree programs and the practical proficiencies sought by employers [3], [4]. We treat that divergence as a quantity to be measured, not as an established fact.

Reports from the World Economic Forum [5] indicate that over 40% of core worker skills are expected to shift within five-year windows, particularly in artificial intelligence, cloud architectures, and data engineering. In addition to specialized technical skills, employer surveys frequently report *perceived* shortfalls in transferable competencies such as professional communication and collaborative problem-solving [6]–[8].

Prior empirical work on curriculum-to-market comparison exhibits four main limitations:

(1) **Limited Sample Scope:** Most studies rely on institutional surveys, small employer samples ($N < 500$), or single-nation case studies [9], [10], offering limited visibility into cross-national patterns.
(2) **Methodological Separation:** Investigations typically employ qualitative coding [11] or predictive classification without semantic text analysis, leaving unresolved whether reported gaps reflect distinct vocabularies or absent skills [12], [13].
(3) **Temporal Relevance:** Several widely cited baseline studies pre-date recent enterprise developments, including widespread cloud migration and the deployment of generative AI tools (2022–2026) [14], [15].
(4) **Limited Theoretical Grounding:** Computational employability models are frequently reported without reference to economic or educational frameworks that would explain *why* curricula and market demands might diverge—frameworks used here to interpret measurements, not to presume their outcome [16], [17].

We address three research questions, each answered against the seven sources that feed the experiments reported here (LinkedIn, HuggingFace, and public benchmark repositories); the wider 15-dataset inventory those sources sit inside, and the reason the gap findings rest on a 1.35M-posting subset of it, is described in Section III and Table I:

- **RQ1 (Market Topography):** What are the most frequent skill requirements in online job postings?

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- **RQ2 (Lexical versus Semantic Alignment):** To what degree do occupational reference-profile descriptors align with market requirements when evaluated using exact string matching versus dense semantic vector embeddings?
- **RQ3 (Domain-Level Reference-Profile Deficits):** What is the magnitude of the reference-profile coverage deficit across distinct subject areas (Programming, Data Science, Cloud Infrastructure, Soft Skills, Cybersecurity)?

Placement-benchmark analyses (employability determinants and cross-cohort placement dynamics; experiments STAT01, STAT02, PLAC01, INTL01) are reported in the companion benchmark-audit article (`paper_placement/`); the geographic distribution of postings (GEO01) is reported within RQ1 above.

Contributions: This work makes five contributions:

1) **Harmonized Multi-Source Corpus:** We assemble the open-data inventory reported in Table I—1.35M real web-mined job postings, a posting-derived occupational reference profile (1,167 postings), 316K unverified aggregated title–skill rows, and 458K benchmark rows of unverified provenance (including 421K student/alumni records; evidence tiers from strong signatures to none found, plus one file declared synthetic by its source, are recorded in `datasets/REAL_VS_SYNTHETIC.csv`).
2) **Dual Lexical–Semantic Analysis:** Using Sentence-BERT (`all-MiniLM-L6-v2`) alongside exact string matching, we show that exact matching yields a lexical SGI of 52.0%—a measurement, not a curricular gap size—whereas semantic vector alignment lowers the estimate to 32.0% at $\tau = 0.5$.
3) **Externally Validated Measurement Instrument:** Beyond in-sample estimates, we validate the matcher against published expert ESCO gold mappings ($\kappa = 0.737$) and re-check both gap indices on a nested-superset robustness sample of 1.29M postings from the same scrape family (a sample-size extension, not an independent replication); a separate computational anchor (embedding coverage, not human audit) relates the occupational reference profile to the CS2023 curricular Body of Knowledge.
4) **Threshold Sensitivity Estimation:** We compute 1,000-fold bootstrap confidence intervals across four similarity thresholds ($\tau \in \{0.4, 0.5, 0.6, 0.7\}$) to evaluate how threshold selection affects gap measurement.
5) **Theory-Informed Interpretation:** Human Capital Theory and Constructive Alignment (Section II-B) are used only to derive the three testable predictions of Section II-D, each evaluated by at least one registered experiment and reported with its verdict in Table VIII; both are interpretive lenses rather than premises of the instrument, so the measurement claims stand independently of any theory. Because this study measures a posting-derived reference profile rather than curricula, it derives no recommendation about course design or assessment; provenance and limitations are documented alongside the measurements.

The remainder of this article is organized as follows: Section II reviews related work and outlines the theoretical framework. Section III describes the datasets, text processing pipeline, and statistical methods. Section IV presents the empirical results for RQ1–RQ3. Section V discusses theoretical implications, an institutional reading of the measurements, reproducibility, and threats to validity. Section VI concludes the study. The placement-model experiments (STAT01, STAT02, PLAC01, INTL01) are audited in a companion article (`paper_placement/`).

II. Related Work and Theoretical Foundations

A. Conceptualizations of the Skills Gap (Literature)

The term *skills gap* is used across macroeconomic, institutional and pedagogical traditions with no settled operational definition—an ambiguity that motivates measuring it explicitly. Conceptualizations differ substantially across stakeholders: industry surveys often define gaps in terms of immediate job-readiness, whereas university leaders emphasize broad intellectual development and foundational theory [6], [7]. Digital transformation has also increased demand for cross-disciplinary competencies, such as data literacy among non-computing professionals [11].

Computational curriculum analytics has sought to quantify these dynamics using course descriptions and accreditation criteria. Cuckov et al. [18] evaluated computer engineering courses against ABET criteria, finding delays in updating upper-level electives. Zhao et al. [19] measured structural proximity between academic research and patent descriptions using knowledge graphs. Focusing on software engineering, Korrapati et al. [20] and Alhazmi et al. [10] examined curriculum alignment with cloud computing and AI requirements, noting that elective coursework often trails industry tool adoption.

B. Theoretical Framework

Two theoretical perspectives inform the interpretation of these empirical observations:

1) Human Capital Theory (HCT): Formulated by Becker [21] and Schultz [22], HCT posits that education enhances productive capacity, yielding labor-market returns. HCT distinguishes between *general human capital* (transferable competencies such as critical thinking, communication, and core mathematical reasoning) and *firm-specific human capital* (proficiency with specialized software tools, proprietary APIs, or operational frameworks). Universities are commonly characterised as prioritizing general human capital because their missions center on enduring principles; we use this as an interpretive lens, not as a premise of the measurement. Conversely, employer job descriptions frequently specify immediately applicable technical tools. This distinction helps explain why measured reference-profile deficits concentrate in applied toolchains such as cloud platforms and data libraries.

2) Constructive Alignment & Curriculum Theory: Biggs [23] introduced constructive alignment to establish coherence among Intended Learning Outcomes (ILOs), teaching methods, and assessment tasks. At an institutional level, any measured divergence can be read through constructive alignment as *external misalignment*—ILOs are governed by multi-year accreditation reviews (such as ABET and ACM/IEEE guidelines), whereas industry skill demands evolve on shorter market cycles. We report this as one interpretation of the measurements, not as a premise of the instrument.

C. Natural Language Processing for Skill-Gap Measurement

Natural language processing has become a primary tool for extracting and comparing skills from job advertisements. Skill identification from postings is itself an established research program, survey-driven since Khaouja et al. [24], and evaluated against human-annotated benchmark corpora whose annotation schemes distinguish skills from qualifications, experience, and occupation [25], [26]. Recent deep-learning surveys catalogue the datasets and the persistent inconsistency in hard/soft-skill terminology that motivates semantic over lexical comparison [27]. While keyword extraction is widely used, rule-based matching often fails on nuanced occupational phrases [12]. Pretrained language representations address this limitation by projecting terms into continuous vector spaces. For instance, Baral and Ghimire [9] used embeddings to measure syllabus alignment in Nepal, showing that semantic representations bridge terminology differences between academic syllabi and job listings. Text-mining methods have similarly been applied to profile data science roles [28], [29] and build resume-matching systems [30]. In this work, we use Sentence-BERT (`all-MiniLM-L6-v2`) [31] to quantify alignment between market requirements and reference-profile descriptors across multiple similarity thresholds, and we validate the resulting matcher against expert mappings rather than assuming its accuracy.

D. Testable Hypotheses

The two theoretical perspectives of Section II-B yield three testable predictions, each evaluated by at least one registered experiment (Table IX):

H1: Constructive Alignment (external misalignment): Because Intended Learning Outcomes are governed by accreditation cycles while toolchains evolve continuously, curricular coverage should be complete for enduring general competencies (soft skills) but incomplete for applied technical toolchains (Cloud, Programming, Data Science) [23]. *Test:* domain deficits in ALIGN03 and their nested-superset re-check in ALIGN04.

H2: Terminology divergence (measurement): If a share of the reported gap is vocabulary mismatch rather than absent competence [12], dense semantic matching should remove a substantial share of the lexically measured gap, and the matches it adds beyond exact string matching should be predominantly valid. Because every exact string match is also a cosine match at $\tau \leq 0.7$, the *direction* of the change is guaranteed by construction; the hypotheses concern the *magnitude* of the reduction (with bootstrap confidence intervals) and the validity of the added matches. *Tests:* ALIGN01 vs. ALIGN02 (magnitude of reduction); ALIGN04 (nested superset) and CURR01 (a third vocabulary) for the same question; VALID01 (expert gold) for the validity of the additional matches.

H3: Instrument robustness: If the Skill Gap Index measures a property of the profile–market pair rather than of one sample of it, both variants of SGI should remain strictly positive and the magnitude of the semantic reduction should persist when the market sample is extended from the 519K-posting primary corpus to a nested superset of the same posting-scrape family. *Test:* ALIGN04 (LinkedIn nested superset, $N = 1.29$M postings).

III. Methodology

A. Data Corpus and Provenance Architecture

The corpus comprises 15 open datasets acquired in September 2026 from Kaggle and HuggingFace repositories under open-access licenses (CC0, CC-BY 4.0, and ODbL). Sources carry per-source observation and release windows listed in Table I; the regional profiles are calibrated against the India Skills Report 2026 (that source declares itself synthetic). Summing raw file row counts yields 3,163,517 records; a provenance audit (`datasets/OVERLAP_AUDIT.csv`) then shows that the two HuggingFace Skill Gap tables share an identical set of 519,814 `job_link` identifiers (a linked pair, not two samples) and that this entire pair is a strict subset of the LinkedIn 2024 postings table (all 519,814 links present), de-duplicating to 2,123,889 unique record rows. That total pools four incompatible row units and is therefore not used as an all-purpose evidence count anywhere in this paper: 1,349,621 real web-scraped posting rows (the 1,348,454 LinkedIn postings plus a separate 1,167-posting occupational-reference file; the 519,814 nested HuggingFace postings contribute no additional rows), 519,814 posting–skill pair rows that vanish on de-duplication, 316,279 aggregated title–skill count rows of unverifiable provenance, and 457,989 benchmark rows of unverified provenance (445,989 signature-flagged—a 10,345-row AI job series with strong generator signatures, a 15,000-row AI salary file with moderate signatures, and 420,644 student/alumni records with moderate-to-weak signatures—plus 12,000 India-skill profiles declared synthetic by their source; evidence tiers are computed from the share of applicable discriminating checks fired, in `datasets/REAL_VS_SYNTHETIC.csv` and `datasets/synthetic_checks_output.json`). Table I reports the same accounting grouped by row unit. The

TABLE I: Dataset Provenance and Structural Inventory (15 sources), grouped by row unit. Raw rows 3,163,517 = 1,869,435 real posting rows + 519,814 posting–skill pair rows + 316,279 unverified aggregate rows + 457,989 unverified-provenance benchmark rows. De-duplication (linked pair + nested corpus) removes $2 \times 519{,}814$ rows, leaving 2,123,889 unique record rows (1,349,621 real posting rows, 316,279 aggregate rows, 457,989 unverified-provenance benchmark rows)—a row total, not a count of distinct postings.

Dataset Identifier	Source Platform & Identifier / Slug	Rows	Domain / Description	Tier	Observation Period	License
Real web-scraped job postings (unit: one posting)						
LinkedIn Jobs & Skills [32]	Kaggle: `asaniczka/1-3m-linkedin-jobs-and-skills-2024`	1,348,454	Global job postings and extracted skill tokens (root of the scrape family)	1	2024	ODbL
HF Skill Gap Jobs	HuggingFace: `skill-gap-analytics/jobs-corpus`	519,814	Enterprise job postings with occupational metadata (strict subset of the LinkedIn postings table; 0 unique rows)	1	2023–2024	CC-BY 4.0
HF Job Skill Set	HuggingFace: `skill-gap-analytics/academic-reference`	1,167	Posting-derived occupational reference profile (a *separate* HuggingFace dataset, not part of the LinkedIn scrape; a posting-level join finds no near-duplicate postings between it and the market corpus—Section III)	1	2023–2024	CC-BY 4.0
Posting–skill pair rows (unit: one posting–skill link; linked twins of the nested postings; 0 unique rows after de-duplication)						
HF Skill Gap Skills	HuggingFace: `skill-gap-analytics/skills-corpus`	519,814	Granular skill requirements linked via `job_id` (same 519,814 IDs as Jobs table)	1	2023–2024	CC-BY 4.0
Aggregated count rows (unit: one title × skill keyword × count row; source scrape unverified)						
USA Job Market Skills [33]	Kaggle: `batuhanklac/usa-job-market-skills-and-title-analysis`	316,279	US title×skill aggregate counts over 6,328 titles (not individual postings)	1	2024	CC0
Unverified-provenance benchmark rows (unit: one listing or profile row; evidence tiers are computed, not assigned; the share of applicable discriminating checks fired (uniformity, coin-flip), cut points ≥0.80 strong, ≥0.50 moderate, ≥0.20 weak/partial, else none found; one file is declared synthetic by its source; `datasets/REAL_VS_SYNTHETIC.csv`, `datasets/synthetic_checks_output.json`)						
AI Trends 2020–2026 [41]	Kaggle: `shree0910/ai-and-data-science-job-market-dataset-20202026`	10,345	AI/data-science job series (benchmark; strong generator signatures)	1	2020–2026	CC0
AI Job Market Trends [42]	Kaggle: `bismasajjad/global-ai-job-market-and-salary-trends-2025`	15,000	AI job and salary rows (benchmark; moderate generator signatures)	1	2024–2025	CC-BY 4.0
India Skills Intelligence [43]	Kaggle: `sanniagarwal88/india-professional-skills-intelligence`	12,000	Profiles declared synthetic by the source README, calibrated on India Skills Report 2026	1	2026 Calibrated	CC0
Student/alumni benchmark records (unit: one student/alumni row; provenance unverified, generator-signature tiers moderate/partial/none found; 0 verified real rows)						
International Graduates [38]	Kaggle: `quackquackrp/international-graduates-employment-dataset`	300,000	Multi-country alumni outcomes benchmark (weak/partial generator signatures)	2	2023–2024	CC0
Student Career Success [37]	Kaggle: `nobeenfatima8/student-career-success-prediction-dataset`	50,000	Student success benchmark (provenance unverified; no generator signatures found)	2	2024	CC-BY 4.0
Career Change Prediction [39]	Kaggle: `jahnavipaliwal/field-of-study-vs-occupation`	38,444	Field-of-study/occupation transition benchmark (moderate generator signatures)	2	2024	CC0
College Placement [34]	Kaggle: `zunesh1507/college-student-placement`	10,000	Placement-classification benchmark (moderate generator signatures)	2	2024	CC0
Global Placement [36]	Kaggle: `rakesh850/global-student-placement-2025-dataset`	10,000	Multi-national placement benchmark (moderate generator signatures)	2	2024–2025	CC0
NGESS Employment [35]	Kaggle: `philipjayo/ngess-graduate-employment-dataset`	7,200	Survey-style graduate cohort (provenance unverified; no generator signatures found)	2	2023–2024	CC0
India Fresher Market [40]	Kaggle: `shambhuraj4jagadale/indian-fresher-job-market`	5,000	Fresher-hiring benchmark rows (weak/partial generator signatures)	2	2024	CC0

occupational-reference file contributes 1,167 job postings rather than a separate 13,833-row taxonomy. Seven of the 15 sources feed the experiments reported below; the remainder (including the generated AI-trends series, retained in the registry but not used here) are retained for inventory completeness (Section V-E). Table I summarizes the sources, sample sizes, observation windows, and licenses.

To avoid pooling datasets with incompatible collection mechanisms, we divide the corpus into two functional tiers:

- **Tier 1: Online Labor Market Postings (1.35M real web-scraped postings; 316K unverified aggregate rows; the linked HuggingFace pair contributes no unique rows):** Comprises the LinkedIn dataset (1.35M records) [32], the USA Job Market dataset (316K records) [33], and the linked HuggingFace Skill Gap jobs–skills pair (519K postings with skill annotations), which the provenance audit shows to be a strict subset of the LinkedIn postings table and which therefore contributes no additional unique records. These corpora contain job advertisements from North America, Europe, and the Asia-Pacific region, providing direct observations of employer skill specifications.
- **Tier 2: Student Placement and Career Benchmarks ($N \approx 421$K student/alumni benchmark rows of unverified provenance, moderate-to-none generator signatures; no verified real graduate records):** Comprises College Student Placement ($N = 10{,}000$) [34], NGESS Graduate Employment ($N = 7{,}200$) [35], Global Student Placement ($N = 10{,}000$) [36], Student Career Success ($N = 50{,}000$) [37], International Graduates ($N = 300{,}000$) [38], Field-of-Study vs. Occupation ($N = 38{,}444$) [39], and India Fresher Market ($N = 5{,}000$) [40] (420,644 records in total). These tabular records provide academic, behavioral, and placement indicators; the placement-benchmark experiments built on them are audited in the companion article (`paper_placement/`).

1) Supplementary Demand-Side Source: ONET 30.0 and BLS OEWS (ALIGN06): Table I inventories the posting corpora behind every headline result. A supplementary demand-side check, experiment ALIGN06, draws instead on two US government sources containing no postings at all. The first is the **ONET 30.0 Database** (release `db_30_0`, August 2025): its *Technology Skills* table (2.B.1.a; 32,681 occupation×technology rows over 1,016 ONET-SOC occupations), plus its *Skills* table (2.A.3.a, Scale ID `IM`) for one weighting variant. O × NET 30.0 Database is used under the **CC BY 4.0** licence: "This analysis uses information from the ONET 30.0 Database by the U.S. Department of Labor, Employment and Training Administration (USDOL/ETA). Used under the CC BY 4.0 license. ONET® is a trademark of USDOL/ETA." The second is the **BLS Occupational Employment and Wage Statistics**, May 2025 national estimates (workbook `national_M2025_dl.xlsx`), supplying employment weights $w(o)$ for 830 detailed occupations: "Source: U.S. Bureau of Labor Statistics, Occupational Employment and Wage Statistics (OEWS), May 2025 national estimates." OEWS is a US federal work in the public domain (17 U.S.C. §105); the BLS emblem is not used. Both archives are pinned by SHA-256 in ALIGN06 `metrics.json`; licence snapshots are archived under `datasets/curriculum/license_snapshots/`.

The profile truncates each ONET-SOC code to its SOC six-digit base, joins that base to the corresponding OEWS occupation, and weights each term by the employment of every occupation listing it (Section IV-B4). It is a *taxonomy-derived* inventory of tools and technologies, not a sample of postings; the market side of every headline result remains the posting corpora of Table I.

B. Taxonomy Construction and Skill Extraction

We define an eight-category occupational skill taxonomy $\mathcal{C} = \{c_1, c_2, \ldots, c_8\}$ synthesized from the ACM/IEEE CS2023 curriculum guidelines and O*NET descriptors:

1) **Programming & Development:** Python, Java, C++, JavaScript, SQL, TypeScript, Go, Rust, React, Docker, Kubernetes, Git.
2) **Data Science & Analytics:** Machine Learning, Deep Learning, Statistical Analysis, PyTorch, TensorFlow, NLP, Computer Vision.
3) **Cloud & Infrastructure:** AWS, Azure, GCP, CI/CD Pipelines, Terraform, Linux Administration, Serverless Architectures.

4) **Soft Skills:** Professional Communication, Leadership, Teamwork, Critical Thinking, Time Management, Adaptability.
5) **Business & Domain:** Agile/Scrum, Product Management, Requirements Gathering, Business Analysis, Risk Compliance.
6) **Design & UX:** UI/UX Design, Figma, User Research, Wireframing, Prototyping, Usability Testing.
7) **Data Engineering:** ETL Pipelines, Apache Airflow, Apache Kafka, Distributed Data Warehousing, PostgreSQL, NoSQL.
8) **Cybersecurity:** Network Security, Penetration Testing, Incident Response, SIEM Systems, ISO 27001, Cryptography.

Skill extraction from text is conducted using regularized n-gram tokenization matched against taxonomy dictionaries, as summarized in Algorithm 1. Occupational skill requirements are represented using a standardized occupational–educational reference profile (job-facing skill lists from five occupational categories) rather than scraped syllabi—a proxy whose content validity is quantified against CS2023 in CURR01 (Section IV-B3); profile-based matching without scraped syllabi follows established practice in semantic resume–job systems [30]. Extraction and matching are fully automated, with matcher validity established externally against expert ESCO annotations rather than in-house re-annotation (Section IV-B1).

Reference-profile circularity check. Because the reference profile and the market corpus are both posting-derived, we checked whether the same postings occur on both sides rather than assuming independence. **Primary, title-free.** An exact set-similarity join over per-posting skill sets compares all 1,167 profile postings against all 515,996 distinct market skill sets (518,953 of the 519,814 primary-corpus postings carry skills) with no title restriction, and finds *no profile/market pairs at skill-set Jaccard* ≥ 0.7 *regardless of title* (maximum 0.667; 0 pairs at ≥ 0.9). **Secondary, title-join.** A normalized-title join followed by a title–skill-set Jaccard comparison finds 0 of the 1,167 profile postings at title–skill Jaccard $\geq$ 0.9; the single collision at Jaccard ≥ 0.5 involves only the generic title "retail sales manager". The looser title-only join matches 575 of 1,167, but these are shared job titles—occupations that exist in both corpora—rather than shared postings, and their skill sets diverge: of the 575, none reaches Jaccard $\geq$ 0.9 and 574 fall below 0.5 (410 between 0.1 and 0.5, 164 below 0.1, and 1 between 0.5 and 0.9). Because a title join can only compare postings whose titles already agree, the title-free join above—which reads no title field at all—is the check that would catch a duplicate whose title differed; the title-join figures are reported as the secondary, more permissive diagnostic. Full counts and the pre-specified leave-out sensitivity design are in `reports/CIRCULARITY_CHECK.md`.

Algorithm 1 Semantic Skill Extraction and Embedding Alignment

Require: Market text corpus $\mathcal{D}_M$, Occupational reference-profile corpus $\mathcal{D}_A$, Taxonomy $\mathcal{C}$, Sentence-BERT model $\mathcal{E}$, Threshold τ
Ensure: Market skill set $\mathcal{M}$, Occupational skill set $\mathcal{A}$, Similarity matrix $\mathbf{S}$, Gap Index $\mathrm{SGI}(\tau)$
$\mathcal{M} \leftarrow \emptyset, \mathcal{A} \leftarrow \emptyset$
for all $d \in \mathcal{D}_M$ **do**
  Extract normalized skill tokens $T_d \leftarrow \mathrm{Tokenize}(d, \mathcal{C})$
  $\mathcal{M} \leftarrow \mathcal{M} \cup T_d$
end for
for all $d \in \mathcal{D}_A$ **do**
  Extract normalized reference-profile tokens $T_a \leftarrow \mathrm{Tokenize}(d, \mathcal{C})$
  $\mathcal{A} \leftarrow \mathcal{A} \cup T_a$
end for
Filter $\mathcal{M}_{\text{top}} \leftarrow$ Top-$K(\mathcal{M}), \mathcal{A}_{\text{top}} \leftarrow$ Top-$P(\mathcal{A})$ $\{K{=}100;\ P{\in}\{100, 500\}$ by experiment (Sec. III)$\}$
Compute dense embeddings: $\mathbf{E}_M \leftarrow \mathcal{E}(\mathcal{M}_{\text{top}}), \mathbf{E}_A \leftarrow \mathcal{E}(\mathcal{A}_{\text{top}})$
Compute pairwise cosine similarity:
$\mathbf{S}_{ij} \leftarrow \frac{\mathbf{e}_{m,i} \cdot \mathbf{e}_{a,j}}{\|\mathbf{e}_{m,i}\|_2 \|\mathbf{e}_{a,j}\|_2} \quad \forall m_i \in \mathcal{M}_{\text{top}}, a_j \in \mathcal{A}_{\text{top}}$
for all $i \in \{1, \ldots, |\mathcal{M}_{\text{top}}|\}$ **do**
  $\sigma_i \leftarrow \max_j \mathbf{S}_{ij}$
  $\mathrm{covered}_i \leftarrow \mathbb{I}(\sigma_i \geq \tau)$
end for
$\mathrm{SGI}(\tau) \leftarrow 1 - \frac{1}{|\mathcal{M}_{\text{top}}|} \sum_{i=1}^{|\mathcal{M}_{\text{top}}|} \mathrm{covered}_i$
return $\mathcal{M}_{\text{top}}, \mathcal{A}_{\text{top}}, \mathbf{S}, \mathrm{SGI}(\tau)$

C. Semantic Vector Alignment via Sentence-BERT

Exact string matching can overestimate profile–market gaps when the occupational reference profile and job postings use different wording for equivalent skills (for example, "interpersonal communication" versus "client communication skills"). To account for semantic similarity, we use the Sentence-BERT model (`all-MiniLM-L6-v2`) [31].

Let $\mathbf{e}_s \in \mathbb{R}^{384}$ denote the normalized dense vector embedding of market skill $s \in \mathcal{M}$, and let $\mathbf{e}_{s'} \in \mathbb{R}^{384}$ denote the embedding of occupational reference-profile descriptor $s' \in \mathcal{A}$. The semantic alignment score is given by:

$$\mathrm{sim}(s, s') = \frac{\mathbf{e}_s \cdot \mathbf{e}_{s'}}{\|\mathbf{e}_s\|_2 \|\mathbf{e}_{s'}\|_2} \tag{1}$$

A market skill s is designated as covered at threshold $\tau \in [0, 1]$ if:

$$\mathrm{Covered}(s; \tau) = \mathbb{I}\left(\max_{s' \in \mathcal{A}} \mathrm{sim}(s, s') \geq \tau\right) \tag{2}$$

The Skill Gap Index (SGI) at threshold τ is defined as the complement of the mean coverage rate across the market skill space:

$$\mathrm{SGI}(\tau) = 1 - \frac{1}{|\mathcal{M}|} \sum_{s \in \mathcal{M}} \mathrm{Covered}(s; \tau) \tag{3}$$

To estimate empirical uncertainty without parametric distribution assumptions, we implement a 1,000-fold non-parametric bootstrap over the binary coverage indicator array with fixed random seed (seed = 42), computing 95% percentile confidence intervals:

$$\mathrm{CI}_{95\%} = \left[\hat{\theta}^*_{(0.025)}, \hat{\theta}^*_{(0.975)}\right] \tag{4}$$

We use $\tau = 0.5$ as the baseline threshold and evaluate sensitivity at $\tau \in \{0.4, 0.5, 0.6, 0.7\}$. The market profile size is fixed at $K = 100$ throughout. The reference-profile covering-set size is $P = 100$ for the headline SGI experiments (ALIGN01/02) and $P = 500$ for the domain-deficit and nested-superset experiments (ALIGN03/04) and CURR01; Section IV-B2 reports a matched-profile sensitivity that shows how SGI level varies with P.

D. Statistical Inference

All interval estimates reported in this article are 95% percentile intervals from the 1,000-fold non-parametric bootstrap defined in Equation 4 (seed 42). Group-comparison statistics for the placement cohort (Welch's t-test, Cohen's d, Cramér's V, and variance-inflation diagnostics) are defined, executed, and reported in the companion benchmark-audit article (`paper_placement/`), together with the classifier and decision-boundary diagnostics; they play no role in the alignment results reported here.

IV. Empirical Results

A. RQ1: Labor Market Skill Topography

Among the 519,814 skill-annotated postings in the HuggingFace corpus, communication and interpersonal competencies appear most frequently (Figure 1); percentages denote the share of postings whose skill string contains the token. The exact token "communication" is specified in 148,765 postings (28.6%), followed by Customer Service (111,342; 21.4%), Teamwork (91,559; 17.6%), "communication skills" (78,182; 15.0%), Leadership (74,517; 14.3%), and Problem Solving (59,713; 11.5%). Near-duplicate variants (e.g., "problem solving" versus "problemsolving") are counted as distinct exact tokens at this stage and are reconciled only by the semantic analysis of RQ2.

Regional parsing of 1.35M LinkedIn records (Figure 2) reveals that the posting corpus is predominantly concentrated in the United States (85.2%), followed by the United Kingdom (8.4%), Canada (4.2%), and Australia (2.2%).

B. RQ2: Lexical vs. Semantic Reference-Profile Alignment

Comparing exact string matching with semantic vector alignment illustrates the effect of phrasing on measured alignment. Under exact string matching, 48 of the Top 100 market skills match the Top 100 occupational reference-profile descriptors, corresponding to a nominal Skill Gap Index of 52.0% (Coverage Rate = 48.0%). Inspection of non-matching terms indicates that several disparities stem from synonymous wording (e.g., "communication skills" versus "interpersonal skills").

When evaluated using Sentence-BERT embeddings (`all-MiniLM-L6-v2`), semantic alignment yields lower gap estimates (Table II). At baseline threshold $\tau = 0.5$, the estimated Skill Gap Index is 32.0% (Coverage Rate = 68.0%, 95% bootstrap CI: [23.0%, 41.0%]).

TABLE II: Sensitivity Analysis: Semantic Vector Threshold (τ) vs. SGI (primary corpus, K=100, P=100)

Threshold τ	SGI	Coverage Rate	95% Bootstrap CI	ΔSGI
$\tau = 0.4$ (Permissive)	0.20	80%	[0.12, 0.28]	-0.12
$\tau = 0.5$ (Baseline)	**0.32**	**68%**	**[0.23, 0.41]**	**+0.00**
$\tau = 0.6$ (Moderate)	0.39	61%	[0.30, 0.48]	+0.07
$\tau = 0.7$ (Strict)	0.46	54%	[0.37, 0.56]	+0.14
Lexical Match (Baseline)	*0.52*	*48%*	*[0.42, 0.63]*	*+0.20*

Evaluated over Top-100 skill distributions. Bootstrap 95% CI via 1,000 resamples (seed 42; experiment ALIGN02). Lexical CI from ALIGN01.

Table III displays illustrative semantic pairings. Terms such as "scheduling" align closely with "time management" (cos = 0.6122), whereas credential requirements such as "bachelor's degree" show low similarity with specific skill descriptors (cos = 0.2937).

TABLE III: Representative Semantic Skill Pairings (`all-MiniLM-L6-v2`)

Market Term	Closest Syllabus Match	Cosine Sim.	Status ($\tau \geq 0.5$)
communication skills	interpersonal skills	0.7355	Covered
problemsolving	problem-solving	0.6396	Covered
scheduling	time management	0.6122	Covered
patient care	customer service	0.5011	Covered
nursing	training	0.4010	Gap
high school diploma	training	0.3682	Gap
bachelor's degree	professionalism	0.2937	Gap

1) External Validation Against Expert Gold Standards (VALID01): Because all alignment metrics rest on the skill matcher, the matcher itself is validated against an independent, human-annotated gold standard rather than in-house labels: the skill-extraction benchmark of [44], whose skill mentions in StackOverflow and public-sector job postings were manually mapped to ESCO skills [26], [45] by domain experts (CC-BY-4.0). The protocol (experiment VALID01; seed 42) forms 1,062 expert-mapped mentions from the TECH and HOUSE test splits into 1,138 positive (mention, skill) pairs and 1,769 negatives—one random unmapped ESCO skill per mention plus sibling-mention skills as in-sentence hard negatives—for 2,907 pairs in total. Lexical rules agree with the expert mappings only slightly (exact match $\kappa = 0.112$, 95% CI [0.091, 0.132]; containment $\kappa = 0.263$), whereas the Sentence-BERT matcher reaches $\kappa = 0.737$ [0.711, 0.760] at the baseline $\tau = 0.5$ (accuracy 87.9% [86.6%, 89.0%], precision 91.9%, recall 75.7%, ROC-AUC of the cosine score 0.961; Table IV). Agreement remains substantial across the operating range (κ from 0.79 at $\tau = 0.4$ to 0.44 at $\tau = 0.7$), and the near-chance lexical κ independently corroborates this section's central finding: exact string matching understates true skill alignment, which is why the semantic SGI is the headline instrument. A complementary vocabulary check against the 13,896-skill ESCO inventory shows that only

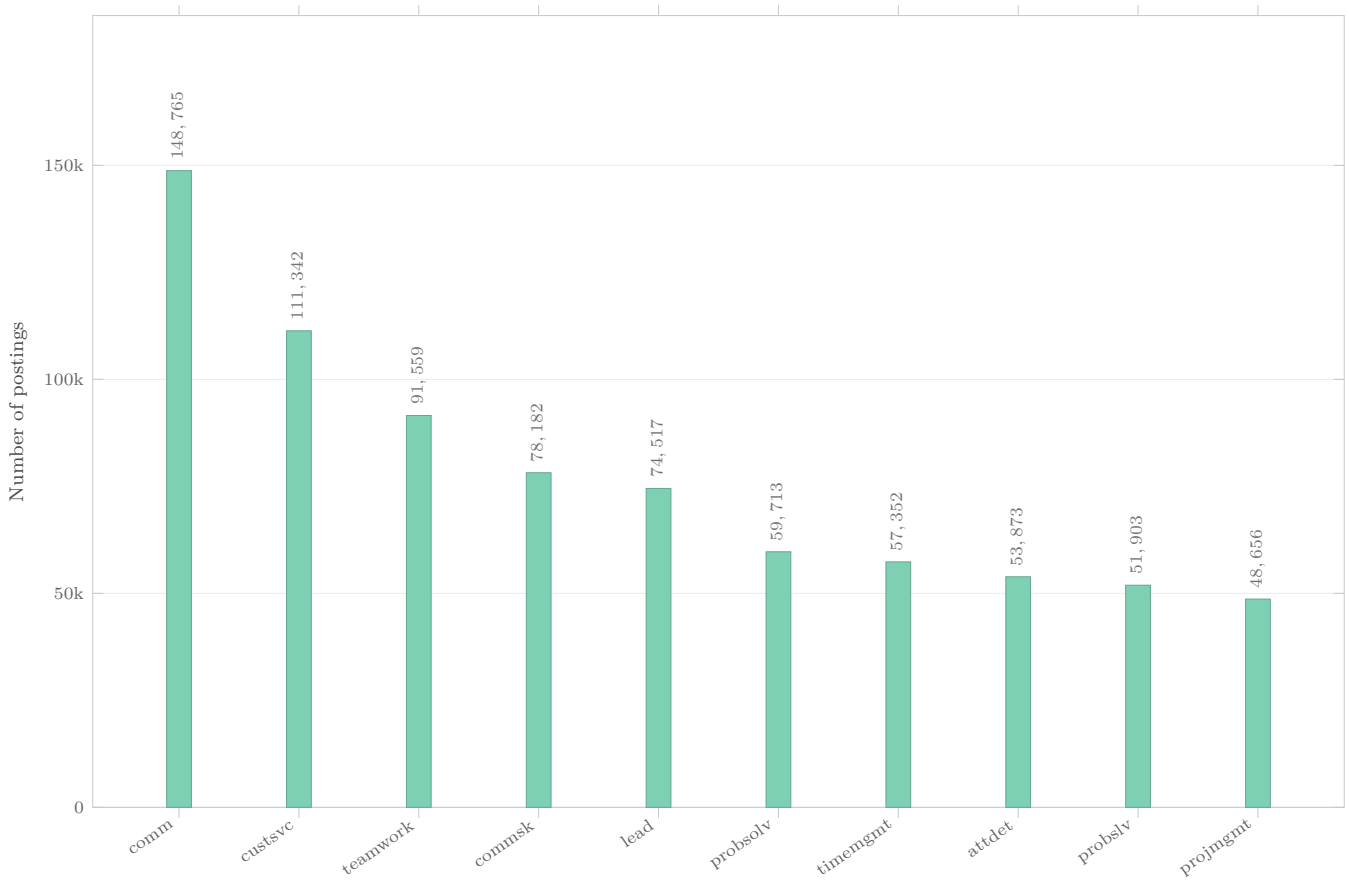


Fig. 1: Top 10 most in-demand exact skill tokens across global postings (N = 519,814 postings; counts are postings containing the exact token; experiment ALIGN01).

6% of the market Top-100 and 11% of the occupational Top-100 appear verbatim in ESCO, yet dense matching at $\tau = 0.5$ resolves 91% [85%, 96%] and 99% [97%, 100%] of them to an ESCO skill under paraphrase—evidence that the extracted vocabularies are semantically grounded in an expert-curated taxonomy.

TABLE IV: Agreement with Expert ESCO Mappings (VALID01; 2,907 Pairs)

Matcher	τ	Acc.	95% CI	Prec.	Rec.	F1	κ	95% CI	AUC
Lexical exact match	—	0.645	[0.628, 0.663]	1.000	0.094	0.172	0.112	[0.091, 0.132]	—
Lexical containment	—	0.696	[0.678, 0.713]	0.941	0.237	0.379	0.263	[0.234, 0.290]	—
SBERT `all-MiniLM-L6-v2`	0.4	0.899	[0.888, 0.910]	0.861	0.885	0.873	0.789	[0.766, 0.812]	0.961
SBERT `all-MiniLM-L6-v2`	0.5	**0.879**	**[0.866, 0.890]**	0.919	0.757	0.830	**0.737**	**[0.711, 0.760]**	0.961
SBERT `all-MiniLM-L6-v2`	0.6	0.829	[0.815, 0.844]	0.957	0.589	0.729	0.614	[0.582, 0.643]	0.961
SBERT `all-MiniLM-L6-v2`	0.7	0.760	[0.744, 0.775]	0.985	0.394	0.562	0.436	[0.405, 0.465]	0.961

Gold: human expert span→ESCO mappings in the SkillSpan-based benchmark of Decorte et al. 95% CIs: 1,000 pair-level bootstrap resamples (seed 42). AUC: ROC-AUC of the cosine score (lexical rules are binary).

2) Nested-Superset Robustness Check on LinkedIn (ALIGN04): A single-sample instrument invites the objection that the measured gap is an artefact of one draw from the market. A pairwise provenance audit shows that the primary 519,814-posting corpus is a strict subset of the LinkedIn 2024 scrape (every `job_link` present), so Experiment ALIGN04 functions as a nested-superset robustness check—a sample-size extension within one scrape family, not an independent replication—by re-running the lexical and semantic procedures on the larger LinkedIn sample (N = 1,294,296 postings with skill annotations), with the occupational covering set at $P = 500$ to match ALIGN03 (top-100 market profile, seed 42, 1,000-fold bootstrap). H2 and H3 receive direct tests under this extension. Exact matching yields SGI = 0.37 [0.28, 0.47] and dense matching at $\tau = 0.5$ yields SGI = 0.14 [0.07, 0.21]; both confidence intervals exclude zero, and the lexical-above-semantic ordering and the monotone increase with τ match the primary-corpus pattern (Table V). Sweeping the full (K, P) grid ($K \in \{50, 100, 200, 500\}$, $P \in \{100, 250, 500, 1000\}$) on this nested superset keeps lexical SGI within $[0.53, 0.74]$ and semantic SGI at $\tau = 0.5$ within $[0.26, 0.38]$, against $[0.52, 0.73]$ and $[0.26, 0.42]$ on the primary corpus—so both ranges are corpus-scoped properties of the grid rather than universal constants.

Profile-size sensitivity. Because ALIGN01/02 use reference-profile $P = 100$ while ALIGN03/04 use $P = 500$, the raw level contrast (HF 0.52/0.32 vs. LinkedIn 0.37/0.14) confounds corpus with covering-set size. We therefore re-estimate SGI on *both* corpora at *both* profile sizes (identical tokenizers, model, and seed): at $P = 100$, HF and LinkedIn both give lexical/semantic SGI = 0.52/0.32; at $P = 500$, both give 0.37/0.14. The instrument is thus exactly stable across corpora at matched

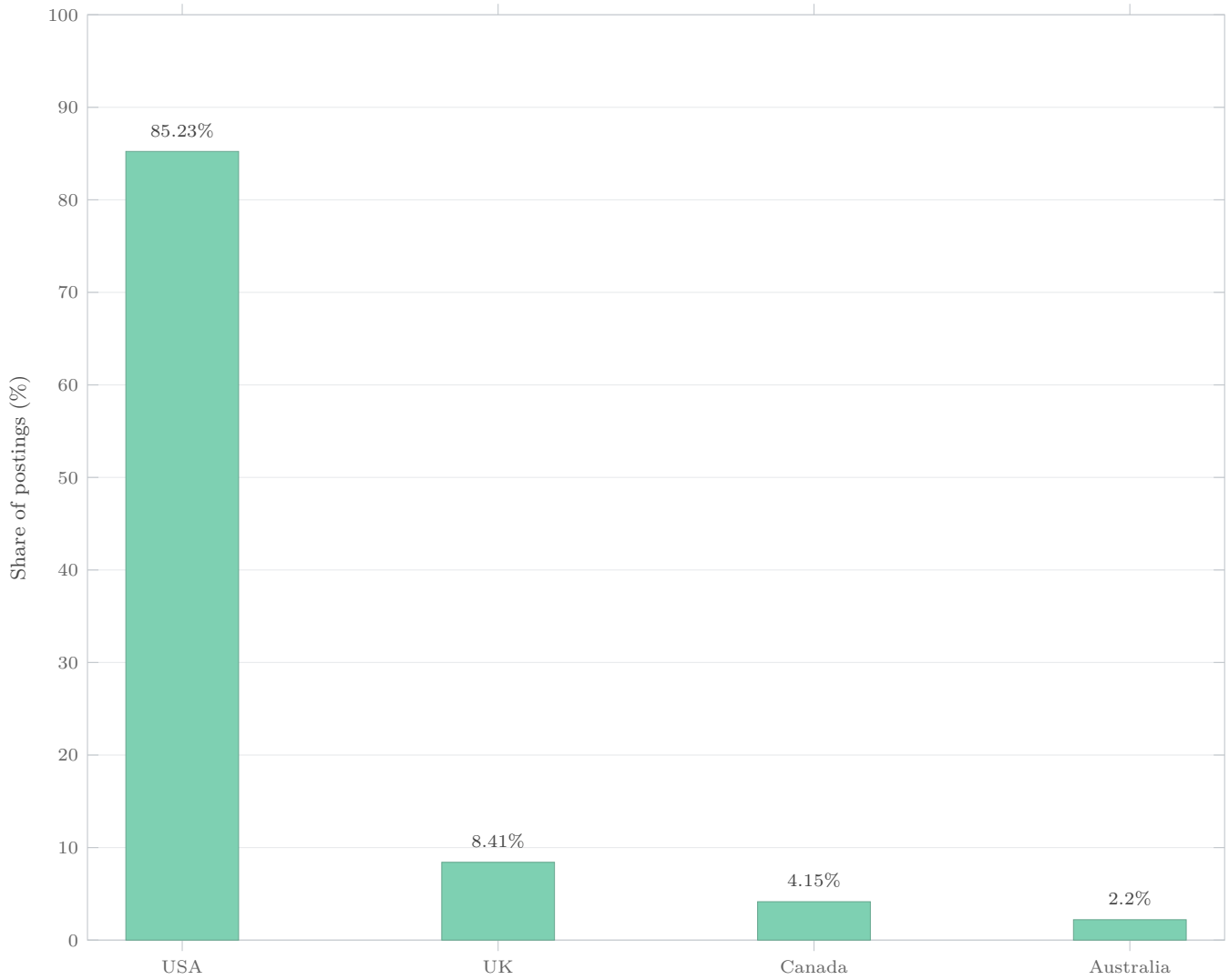


Fig. 2: Geographic concentration of 1.35M job postings (GEO01).

profile size—supporting H3 even more cleanly than a raw level comparison—while the *level* of SGI is a property of the reference-profile covering-set size, not of the market corpus. We therefore report all SGI values with their (K, P) settings and with corpus-specific confidence intervals rather than as universal constants; the earlier vocabulary-composition explanation of the level shift is withdrawn in favour of this measured profile-size sensitivity.

TABLE V: Nested-Superset SGI Check (ALIGN01/02 vs. ALIGN04)

Matcher	Occupational $P = 100$ HF	Occupational $P = 100$ LinkedIn	Occupational $P = 500$ HF	Occupational $P = 500$ LinkedIn
Lexical exact	**0.52 [0.42, 0.63]**	0.52 [0.42, 0.62]	0.37 [0.28, 0.47]	**0.37 [0.28, 0.47]**
SBERT τ=0.5	**0.32 [0.23, 0.41]**	0.32 [0.23, 0.42]	0.14 [0.08, 0.21]	**0.14 [0.07, 0.21]**

Top-100 market skills vs. occupational covering sets on each corpus (ALIGN01/02: P=100; ALIGN04: P=500; matched-profile sensitivity in Section IV-B2). 95% CIs: 1,000 bootstrap resamples (seed 42). Bold: baseline $\tau = 0.5$.

3) Reference-Profile Anchor Validation against CS2023 (CURR01): The occupational reference profile remains a proxy for curricula (Section V-E, item 3). To anchor it, experiment CURR01 compares the profile against the authoritative CS2023 curricular Body of Knowledge—the joint ACM/IEEE-CS/AAAI curriculum guidelines [46]—whose 2,197 topic statements across 17 Knowledge Areas define what computing programs are expected to teach. Semantic coverage of CS2023 topics by the reference-profile top-500 is 30.4% [28.5%, 32.4%] at $\tau = 0.5$ (63.1% at the permissive $\tau = 0.4$), and the reverse—the share of profile skills grounded in some CS2023 topic—is 49.6%. Coverage varies sharply by Knowledge Area (Table VI): Human-Computer Interaction (52.9%) and Data Management (52.5%) are best represented, while Algorithmic Foundations (12.6%), Mathematical and Statistical Foundations (10.0%), and Graphics & Interactive Techniques (11.6%)—the formal and theoretical core of the discipline—are least represented. Exact lexical coverage is near zero for every Area (0–2.4%), consistent with H2 on a third, genuinely independent vocabulary (curricular guidelines, not postings). The anchor is

computational—embedding coverage, not a human line-by-line topic audit—so it bounds the profile without converting coverage into a curricular quantity. It thus both bounds and partially validates interpretation: the reference profile partially reflects the *applied* component of computing curricula that is visible in job-facing skill descriptions (roughly half of applied Knowledge-Area topics at $\tau = 0.5$), but it should not be read as representing the full theoretical core—a limitation now quantified per Area rather than asserted.

TABLE VI: Semantic Coverage of CS2023 Knowledge Areas by the Occupational Reference Profile (CURR01; $\tau = 0.5$)

KA	Knowledge Area	n topics	Cov.%	95% CI
HCI	Human-Computer Interaction	85	52.9	[42.4, 63.5]
DM	Data Management	137	52.5	[44.5, 61.3]
SE	Software Engineering	168	44.0	[36.3, 51.2]
NC	Networking and Communication	65	41.5	[30.8, 55.4]
SEP	Society, Ethics, Profession	161	41.0	[32.9, 48.4]
SF	Systems Fundamentals	83	36.1	[26.5, 47.0]
SPD	Specialized Platform Dev.	201	34.3	[27.9, 40.8]
AI	Artificial Intelligence	215	32.6	[26.1, 38.6]
OS	Operating Systems	148	31.8	[24.3, 38.5]
SDF	Software Dev. Fundamentals	58	31.0	[20.7, 43.1]
SEC	Security	103	30.1	[21.4, 39.8]
PDC	Parallel and Distributed Comp.	60	23.3	[13.3, 35.0]
FPL	Foundations of Prog. Languages	295	18.3	[13.9, 22.7]
AR	Architecture and Organization	111	13.5	[7.2, 19.8]
AL	Algorithmic Foundations	103	12.6	[5.8, 19.4]
GIT	Graphics and Interactive Tech.	164	11.6	[7.3, 16.5]
MSF	Math. and Statistical Found.	40	10.0	[2.5, 20.0]

Coverage: share of that Area's CS2023 topics with $\max \cos \geq 0.5$ to the reference-profile top-500 (1,000 topic-level bootstrap resamples, seed 42). Sorted descending; lexical coverage is 0–2.4% throughout.

4) Supplementary Demand-Side Check against ONET × BLS Employment (ALIGN06): Every result above places a posting-derived market profile against a posting-derived reference profile; a remaining objection is that both sides are the same kind of text. Experiment ALIGN06 rebuilds the market profile from the ONET 30.0 *Technology Skills* taxonomy weighted by BLS OEWS May 2025 employment (Section III-A1), so the demand side contains no postings, while the reference profile, matcher, K grid, bootstrap design and seed 42 are unchanged. It is a diagnostic of the instrument, not an independent confirmation of the gap. *Estimand caveat, carried throughout:* the ONET side is employment-weighted occupations rather than postings, and skill-level SGI under this pairing is vocabulary-dominated rather than demand-driven. The two vocabularies intersect in 90 of the 8,613 ONET terms (1.0%), and top-K Jaccard against either posting corpus is 0.000–0.012. At that degree of overlap a positive SGI is close to guaranteed: an exact matcher cannot match a term the demand vocabulary carries and the reference vocabulary lacks, so the index is driven by inventory size rather than by mismatch in demand. The levels are consequently not comparable across demand sources—SGI = 0.97 [0.93, 1.00] lexical and 0.68 [0.59, 0.77] at $\tau = 0.5$ against the posting-based 0.52 and 0.32 measures that vocabulary difference, not a larger gap. ALIGN06 is not cited here as corroboration, and no headline number is edited from it.

The same limitation persists one level up, so the eight-category comparison is a descriptive diagnostic and not a comparable quantity either. The ONET *Technology Skills* list carries no soft-skill, business or cybersecurity demand weight (0.00 pp each, against 52.26 pp, 11.81 pp and 0.62 pp in postings), and the largest category differs: Programming & Development in ONET at 21.67 pp, against Cloud & Infrastructure in postings at the largest lexical deficit, 2.45 pp. ONET demand concentrates in Programming & Development (21.67 pp), Data Science & Analytics (14.87 pp), Cloud & Infrastructure (13.21 pp) and Data Engineering (9.83 pp), where exact matching covers none and, at $\tau = 0.5$, semantic coverage reaches 92.1% for Cloud & Infrastructure (deficit 1.04 pp [0.00, 2.68]) and 51.9% for Data Engineering (4.73 pp [2.92, 13.33]); a further 39.1% of employment-weighted demand falls outside the taxonomy, so the ONET side constrains technical tool coverage only. Table X reports both sources side by side, and the full (K, P, τ) grid, the weighting variants, the cross-source overlap statistics and the occupation-to-employment join multiplicity are in Appendix A.

C. RQ3: Domain-Level Skill Gap Quantification

For each of the eight taxonomy domains we quantify: (i) *market demand*—the share of postings whose skill set requires at least one taxonomy skill of the category; (ii) *coverage rate*—the occurrence-weighted share of those category-skill requirements present in the occupational top-500 reference profile, evaluated by exact match and by dense matching at $\tau = 0.5$; and (iii) the resulting *deficit* in percentage points of postings, deficit = demand × (1 − coverage). Confidence intervals come from a 1,000-fold non-parametric bootstrap over postings (seed 42; experiment ALIGN03). Because the posting corpus is dominated by generic and non-technical competencies (Section V-E), absolute taxonomy deficits are small in magnitude; severity is therefore assigned *relative to the cross-category median deficit* (Critical ≥ 2× the median, Moderate ≥ median, Mild otherwise).

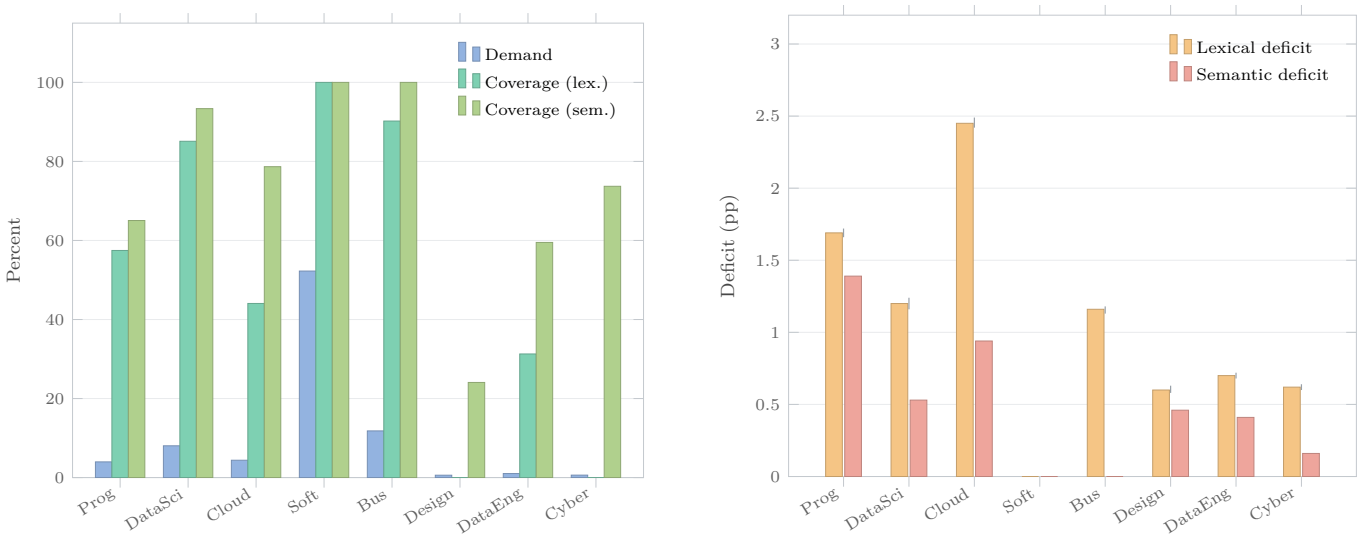
Table VII and Figure 3 summarize the results.

TABLE VII: Domain-Level Skill Gaps Across Taxonomy Categories (ALIGN03)

Category	Demand (%)	Lexical Cov. (%)	Lexical Deficit	Semantic Cov. (%)	Semantic Deficit	Sev.
Programming & Development	3.97	57.5	1.69 [1.66, 1.72]	65.0	1.39	Moderate
Data Science & Analytics	8.04	85.1	1.20 [1.16, 1.24]	93.4	0.53	Moderate
Cloud & Infrastructure	4.39	44.1	2.45 [2.42, 2.49]	78.7	0.94	**Critical**
Soft Skills	52.26	100.0	0.00 [0.00, 0.00]	100.0	0.00	Mild
Business & Domain	11.81	90.2	1.16 [1.13, 1.18]	100.0	0.00	Moderate
Design & UX	0.60	0.0	0.60 [0.58, 0.63]	24.1	0.46	Mild
Data Engineering	1.02	31.3	0.70 [0.68, 0.72]	59.5	0.41	Mild
Cybersecurity	0.62	0.0	0.62 [0.60, 0.64]	73.7	0.16	Mild

Demand: % of postings requiring ≥ 1 category skill. Deficit: pp of postings, 95% bootstrap CI. Coverage: share of category-skill occurrences in the reference profile (lexical; semantic at $\tau = 0.5$). Severity: relative-to-median rule.

Soft Skills dominate market demand (52.3% of postings require at least one soft skill) and are fully covered by the reference profile (deficit 0.00 pp). Among technical categories, Cloud & Infrastructure exhibits the largest deficit (2.45 pp, 95% CI [2.42, 2.49]; the only Critical classification): only 44.1% of its skill requirements appear



(a) Market demand vs. reference-profile coverage

(b) Deficit magnitude (pp) with 95% CI

Fig. 3: Skill-gap analysis across taxonomy categories (ALIGN03). (a) Market demand versus coverage by the occupational reference profile (lexical and dense semantic at $\tau = 0.5$). (b) Deficit in percentage points of postings with 95% bootstrap confidence intervals. Cloud Infrastructure is the only Critical deficit under the relative-to-median severity rule.

in the reference profile lexically, though dense semantic matching recovers coverage to 78.7% and reduces the deficit to 0.94 pp. Programming & Development shows the second-largest lexical deficit (1.69 pp; 57.5% coverage), followed by Data Science & Analytics (1.20 pp despite 85.1% lexical coverage) and Business & Domain (1.16 pp). Semantic alignment reduces the residual reference-profile deficit in every category with a nonzero deficit—by 74% in Cybersecurity, 62% in Cloud, 56% in Data Science, and 41% in Data Engineering—and eliminates the Business & Domain deficit entirely, consistent with vocabulary divergence accounting for a substantial share of the raw lexical gap. Design & UX and Cybersecurity have zero lexical coverage but occur in under 1.1% of postings each, limiting their aggregate impact to below 0.7 pp.

The directional structure of H1 persists on the nested LinkedIn superset (ALIGN04): the same category ordering emerges with Programming & Development carrying the largest semantic deficit (1.39 pp [1.38, 1.41]) and Soft Skills again at 0.00 pp, while absolute magnitudes are smaller—consistent with two corpus-composition facts. First, free-text LinkedIn skills under-detect exact taxonomy keywords, so category demand shares are conservative; second, the same generic-term dominance documented for the SGI lowers technical coverage rates structurally. This check therefore supports the ordering of H1 while indicating that deficit magnitudes are corpus-conditional (Section V-E, item 7).

V. Discussion

A. Theoretical Implications

Hypothesis verdicts. Table VIII summarizes the three predictions derived in Section II-D. H1, a constructive-alignment prediction about curricular coverage, is *consistent with* the measurements on the primary corpus and persists on the nested LinkedIn superset (one scrape family): technical categories carry the only nonzero reference-profile deficits while soft skills are fully covered. Because the instrument measures a posting-derived profile rather than curricula, we report consistency, not confirmation of the curricular claim. H2 (terminology divergence) is supported by the primary ALIGN01/02 contrast (0.52 vs. 0.32), echoed within the same scrape family by the nested ALIGN04 check (0.37 vs. 0.14), and independently attested on curricular vocabulary by CURR01 (lexical 0–2.4% vs. semantic 30.4%)—and H3 (instrument robustness) is supported by strictly positive SGI and magnitude persistence on the nested LinkedIn superset (the lexical-above-semantic ordering is retained only as a sanity check, since every exact string match is also a cosine match of $1.0 \geq \tau$ by construction), and more strongly still by the matched-profile sensitivity showing identical SGI on both nested samples of the scrape family at equal reference-profile covering-set size. Placement hypotheses H4–H6 and their benchmarks (STAT01, STAT02, PLAC01, INTL01) are evaluated as a benchmark audit in the companion article (`paper_placement/`).

TABLE VIII: Theory-Derived Hypotheses and Verdicts

Hyp.	Source theory	Verdict (experiment)
H1	Constructive-alignment prediction	Consistent with measurements (ALIGN03, ALIGN04)
H2	Terminology divergence	Supported ×3 (ALIGN01/02, ALIGN04, CURR01)
H3	Instrument robustness	Supported (ALIGN04)

1) Terminology Divergence and Semantic Alignment: Studies evaluating curriculum alignment through exact keyword matching often report misalignment rates

exceeding 50% [9]. Our findings qualify this observation. While exact string matching yields an apparent gap of 52.0%, dense vector embeddings reduce the estimated deficit to 32.0% at $\tau = 0.5$ (95% CI: [23.0%, 41.0%]). This divergence indicates that a portion of the measured gap arises from vocabulary differences: occupational reference profiles and curricular descriptors often use conceptual terms (e.g., interpersonal communication or database principles), whereas job postings emphasize specific tool names or operational tasks. That expert-annotated lexical agreement is near chance ($\kappa = 0.112$; VALID01) while semantic agreement is substantial ($\kappa = 0.737$) shows this is a measurement property of keyword methods, not a modeling preference.

B. Institutional Factors: One Interpretation of the Measured Deficits

The persistent 32.0% semantic deficit, and the domain-level deficits concentrated in Cloud Infrastructure (2.45 pp lexical; 0.94 pp semantic) and Programming & Development (1.69 pp lexical), are consistent with—though do not by themselves establish—institutional constraints in curriculum adaptation [23], reported here as an interpretation of the measurements rather than as a finding:

1) **Curriculum Governance and Accreditation:** Formal degree reviews and accreditation bodies operate on multi-year cycles (typically 3 to 6 years)—a documented institutional mechanism offered as an interpretation of the measurements. Modifying course requirements or introducing new technical electives requires administrative and faculty committee approvals that may lag industry tool adoption.
2) **Faculty Development Latency:** Teaching rapidly evolving frameworks (e.g., container orchestration, distributed cloud platforms, specific ML toolchains) requires ongoing technical training, which institutional development budgets rarely prioritize over core research obligations.
3) **Infrastructure and Computing Costs:** A candidate explanation this study does not test is that maintaining cloud accounts, GPU-capable compute, and industry-grade data environments imposes financial and operational constraints on computer science and engineering departments; establishing it would require cost and infrastructure evidence, which this sensitivity audit does not collect.

C. Implications of the Measurement Results

The measurements support claims about the instrument and the corpus—which vocabularies an exact matcher can and cannot resolve, how much of the posting-derived skill space the eight-category taxonomy covers, and how far a posting-derived reference profile sits from a technology inventory. They support no claim about curricula: no curriculum was read at course level and no institution is observed acting on any measurement, so this study derives no recommendation about course design, assessment practice, or accreditation. Four open questions follow from the measurements, and this study cannot answer any of them. Whether the 44.1% lexical coverage of Cloud Infrastructure reflects curriculum content or the vocabulary of a keyword matcher applied to posting text would require course-level reading and human annotation of curricula. Whether the concentration of non-zero deficits in technical categories is an effect of corpus composition—the posting corpus is dominated by generic and non-technical competencies—rather than of curriculum content is untested here. Whether the deficits persist outside this single scrape family, and outside a corpus that is 85.2% United States, is bounded only by the nested-superset check of ALIGN04, which is a sample-size extension within the same family. Whether a different taxonomy or a different covering-set size P would move the deficits as much as the matcher threshold does is likewise unmeasured. These are recorded as open questions rather than as implications for practice, which is the scope Contribution 5 states.

D. Reproducibility and Data Availability

Every quantitative result reported in this article comes from a registered experiment and is stored as a machine-readable artifact under `results/`, from which the tables and figures are generated rather than transcribed by hand. An automated, read-only check (`audit/verify_claims.py`) compares each manuscript quantitative constant against those stored artifacts (103 checks, agreement required within 10^{-6}); it verifies that the paper reports what the artifacts contain, and is explicitly distinct from re-deriving those artifacts from the raw data. Separately, as a stronger and independent check, all 13 registered experiments were re-executed end-to-end on CPU-only machines with pinned dependencies, fixed seeds and single-threaded arithmetic: the 8 that load no embedding model ran offline on one machine (288 compared numeric fields; `reports/CPU_VS_COMMITTED.md`), and the 5 that load Sentence-BERT—which a local application-control policy blocks from running on that machine—ran on a CPU-only Kaggle kernel against the same pinned input bundle (614 compared numeric fields; `reports/CPU_VS_COMMITTED_KAGGLE.md`). Across the two runs, headline metrics agree at reported precision; for the 5 Kaggle experiments the per-pair cosine similarities stored in the CSV artifacts differ by at most 10^{-4} (CPU vs GPU float differences), so their agreement is not claimed as bit-exact, whereas the 8 locally re-run experiments reproduce bit-exactly. All 902 compared `metrics.json` numeric fields reproduced exactly, every difference being zero rather than merely within the 10^{-6} relative tolerance, and the Kaggle outputs are archived with the others (`anon/skill-gap-cpu-verification`). Table IX lists the experiment identifiers reported here and their verification status; the retained-but-unused TRENDS01 entry is held in the supplement table

TABLE IX: Experiment Registry: Provenance of All Reported Results (status per `results/registry.json`)

Experiment ID	Task	Att.	Status
ALIGN01	Lexical alignment + SGI bootstrap (ALIGN01)	1	USED_IN_MANUSCRIPT
ALIGN02	SBERT semantic alignment, τ sweep (ALIGN02)	3	USED_IN_MANUSCRIPT
ALIGN03	Eight-category domain gap (ALIGN03)	2	USED_IN_MANUSCRIPT
GEO01	Geographic shares (GEO01)	1	USED_IN_MANUSCRIPT
VALID01	External gold-standard matcher validation (VALID01)	1	USED_IN_MANUSCRIPT
ALIGN04	Nested-superset SGI re-check, LinkedIn (ALIGN04)	4	USED_IN_MANUSCRIPT
CURR01	CS2023 curriculum anchor validation (CURR01)	3	USED_IN_MANUSCRIPT
ALIGN06	O*NET × BLS employment-weighted demand profile, supplementary (ALIGN06)	2	USED_IN_MANUSCRIPT

Each experiment receives an immutable input bundle from a public dataset (`anon/skill-gap-inputs-v1`; the owning namespace is withheld for double-blind review) and writes `metrics.json` plus CSV artifacts; outputs are downloaded, schema- and sanity-verified (`experiments/verify.py`), and only `VERIFIED` outputs are used in this manuscript. Table and figure generation is automated by `experiments/make_latex.py` from those verified artifacts.

(`paper/tables/tab_experiments_supplement.tex`); the placement-model experiments (STAT01, STAT02, PLAC01, INTL01) are reported in the companion benchmark audit (`paper_placement/`). Inputs are listed with licence and SHA-256 checksum in `datasets/DATASET_MANIFEST.csv` and `datasets/input_checksums.sha256`, and the step-by-step clean-room procedure is in `reports/CLEAN_ROOM_REPRO.md`.

The analyses were implemented in Python 3.14.6 with `scikit-learn` 1.9.0, `statsmodels` 0.15.0, `sentence-transformers` 6.0.1 (`all-MiniLM-L6-v2`), `scipy` 1.18.1, `pandas` 3.0.5, and `numpy`; stratified cross-validation splits, bootstrap resamplings, and all estimators use fixed seeds (seed = 42; multi-seed robustness uses seeds 0–9). Complete experiment outputs are archived in a public HuggingFace dataset (`anon/skill-gap-experiments`; namespace withheld for double-blind review), the runnable experiment suite and scheduler live in the project repository (`experiments/`), and all 15 source datasets are publicly accessible under open-access licenses (CC0, CC-BY 4.0, ODbL; Table I and `DATASET_MANIFEST.csv`).

E. Threats to Validity and Limitations

Several methodological constraints should be considered when interpreting these findings:

1) **Data Provenance & Benchmark Artifacts:** While Tier 1 datasets contribute 1.35M unique web-mined postings (the HuggingFace Skill Gap tables are a strict subset of the LinkedIn scrape rather than a second sample; Section IV-B2), the Tier 2 student benchmarks are of unverified provenance (moderate-to-none generator signatures; `datasets/synthetic_checks_output.json`) and are excluded from the gap findings here; their provenance, decision-boundary, and transfer audit is reported in the companion benchmark article (`paper_placement/`). The occupational reference profile is drawn from 1,167 job postings in five occupational categories rather than from institutionally collected syllabi.
2) **Curriculum Proxy Limitations:** Academic skill demands were proxied through normalized occupational and educational reference profiles rather than direct, institutionally mined university syllabi across diverse global departments. Experiment CURR01 anchors this proxy to the CS2023 Body of Knowledge [46], quantifying both its validity (30.4% topical coverage, 49.6% of profile skills grounded in the guidelines) and its bound. The anchor is a computational embedding-coverage measure—no human line-by-line topic audit was performed—so neither figure converts into a curricular quantity. Coverage is concentrated in applied Knowledge Areas (HCI, Data Management, Software Engineering) and weak over the theoretical core (Algorithmic Foundations 12.6%, Mathematical Foundations 10.0%), so conclusions about applied skill coverage are anchored while claims about the full curricular core are not.
3) **Reference Profile Is Not a Curriculum Sample:** Every deficit reported here is the difference between a market corpus and a posting-derived reference profile. No university syllabus, course document or programme specification was collected or analysed, so these measurements do not establish the size—or the existence—of a curricular gap; CURR01 bounds the profile against CS2023 but does not convert the result into a curricular quantity.
4) **Scope of Human Validation:** Matcher validity is established against published expert gold standards (SkillSpan→ESCO mappings; $\kappa = 0.737$ at $\tau = 0.5$; Section IV-B1) rather than in-house annotation of this specific corpus, and instrument robustness is supported by a nested-superset check (Section IV-B2); independent-rater annotation of the present skill pairs remains future work.
5) **Embedding Model and Threshold Sensitivity:** While Sentence-BERT mitigates surface vocabulary divergence, estimated alignment remains sensitive to cosine similarity cutoff selection (SGI = 20.0% at $\tau = 0.4$ versus 46.0% at $\tau = 0.7$) and to reference-profile covering-set size (P): the matched-profile sensitivity in Section IV-B2 shows SGI levels shifting from 0.52/0.32 ($P = 100$) to 0.37/0.14 ($P = 500$) on both nested corpora of the scrape family, so cross-experiment SGI comparisons are valid only at equal (K, P).

6) **Geographic Concentration:** 85.2% of the LinkedIn posting corpus originates from the United States, skewing technical demand profiles toward Western, mature tech economies.
7) **Corpus Composition & Conditional Demand:** The 519K-posting skills corpus is dominated by generic, healthcare, and retail competencies (e.g., BLS/acls credentials and nursing appear among the Top-100 market tokens). Consequently, taxonomy-technical demand shares are modest (e.g., 3.97% of postings require at least one Programming skill), and the RQ3 deficits should be read as *coverage* gaps conditional on demand rather than as economy-wide demand shares. The nested LinkedIn check (ALIGN04) shows both sides of this conditioning: deficits shrink further on free-text skills (Programming 1.39 pp), and exact keyword taxonomy matching under-detects demand in corpora whose skill strings are not pre-normalized—so comparisons across the two annotation layers are valid for the *ordering* of deficits but not their absolute magnitudes.
8) **Automated Severity Calibration:** Because absolute domain deficits are small, the Critical/Moderate/Mild labels use a relative-to-median rule rather than externally anchored thresholds; different anchors would reassign labels, though the ranking of categories is invariant.
9) **Overlap Detection Relies on Title Matching:** The reference profile carries no shared posting identifier with the market corpus, so the posting-level overlap check must fall back to normalized-title matching, which is a necessary rather than sufficient condition: generic titles such as "Human Resources Manager" recur hundreds of times in the market corpus, so title agreement alone is not evidence of a shared posting and the check requires skill-set corroboration. A duplicate whose title happened to differ from every profile title would be invisible to that join; to bound this, a title-free exact set-similarity search over per-posting skill sets (1,167 profile postings against 515,996 distinct market skill sets) finds a maximum Jaccard of 0.667 and no pair at ≥ 0.7, so undetected skill-level duplication does not appear to be material. The overlap estimate is nonetheless a bound under a title-based join rather than a direct identifier match. Scope: the title-free search covered the primary 519,814-posting market corpus (518,953 postings carrying skills); the 1,294,296-posting superset used by ALIGN04 was not searched, so the 0.667 bound attaches to the corpus behind the headline indices, not to the robustness superset.
10) **The Demand-Side Check Is Not an Independent Confirmation (ALIGN06):** The supplementary ONET × BLS check (Section IV-B4) must not be read as corroborating the gap. Skill-level SGI under this pairing is vocabulary-dominated: the demand and reference vocabularies share 90 of 8,613 ONET terms (1.0%) and top-K Jaccard against the posting corpora is 0.000–0.012, so a positive SGI is close to guaranteed and its level (0.97/0.68) is not comparable to the posting-based 0.52/0.32. The demand side is employment-weighted occupations rather than postings. Aggregating onto the eight taxonomy categories does not repair this either: the technology inventory maps no demand weight to Soft Skills, Business & Domain or Cybersecurity (0.00 pp each, against 52.26, 11.81 and 0.62 pp in postings), the largest category differs across sources (Programming & Development 21.67 pp in ONET against Cloud & Infrastructure at a 2.45 pp lexical deficit in postings), and 39.1% of employment-weighted demand falls outside the taxonomy, so category-level results are also not comparable across demand sources. No headline figure is revised from ALIGN06.

VI. Conclusion

On the tiered corpus described in Section III (1.35M real web-mined postings, with 316K unverified aggregate rows and 458K unverified-provenance benchmark rows recorded but excluded from these findings), this empirical investigation—which proxies academic skill requirements with an employer-derived occupational reference profile rather than university syllabi—shows that profile–market misalignment is twofold. A substantial share of the apparent deficit reflects vocabulary divergence between reference-profile descriptors and recruitment postings (the SGI falls from 52% to 32% under semantic alignment, and several domain deficits shrink by more than half), while substantive residual reference-profile deficits remain concentrated in rapidly evolving technical areas, specifically Cloud Infrastructure and Programming & Development. Whether those residuals correspond to curricular shortfalls is not established by this study. The instrument supporting this claim is examined three ways: expert-gold agreement on published ESCO mappings ($\kappa = 0.737$), a nested-superset robustness check on 1.29M postings from the same scrape family (both indices strictly positive; identical SGI at matched reference-profile covering-set size), and a computational anchor to the CS2023 curricular guidelines (embedding coverage concentrated in applied knowledge areas; no human line-by-line topic audit was performed). If the residual reference-profile deficits do correspond to curricular shortfalls, addressing them would not necessitate wholesale curriculum redesign—an inference we mark as conditional and do not test here. Because the measurements cannot by themselves separate vocabulary divergence from curricular shortfall, the next step is a study that measures curricula directly; the modular-credential and assessment options in Section V remain candidate responses conditional on such evidence.

Appendix A
ALIGN06 Grid, Cross-Source and Join Detail

This appendix carries the numerical detail of the ALIGN06 demand-side check summarised in

Section IV-B4. The attribution in Section III-A1 and the estimand caveat in Section IV-B4 remain in the main text and apply to every value below. All values come from `results/ALIGN06_onet_demand_profile/`, seed 42, 1,000-fold bootstrap.

Profile composition. The most ubiquitous economy-wide term is "microsoft office software", listed by 94% of frame occupations. The ONET *Technology Skills* table is a software-and-hardware inventory rather than a job-skill taxonomy, which is the proximate cause of the vocabulary domination described in Section IV-B4.

Grid. Over the full (K, P) grid, economy-wide lexical SGI stays within $[0.97, 0.985]$ and semantic SGI at $\tau = 0.5$ within $[0.57, 0.72]$. Restricting the frame to the 18 computing/IT occupations (13 SOC bases) gives 0.97 lexical and 0.72 semantic at $\tau = 0.5$. The five weighting-and-vocabulary variants at $K = 100$, $P = $ all, $\tau = 0.5$ span $[0.60, 0.72]$: commodity-title 0.60, importance-weighted 0.66, unweighted and economy-wide 0.68, computing/IT 0.72. At $K = 100$, $P = $ all, SGI increases monotonically with τ—0.43, 0.68, 0.80, 0.89 for $\tau \in \{0.4, 0.5, 0.6, 0.7\}$—and the skill-level bootstrap interval excludes zero at every τ, which is the condition under which the preregistered persistence claim is permitted.

Cross-source overlap. Top-K Jaccard between the ONET profile and the two posting corpora is 0.000–0.012 for $K \in \{50, 100, 200, 500\}$, and the Spearman correlation of the two weight vectors is approximately -0.81 to -0.83. Both statistics are descriptive only and were never used to select parameters. The strongly negative rank correlation is a mechanical consequence of near-disjoint vocabularies—most vector mass sits on terms absent from the other profile—and is not evidence that the two demand sources disagree on a common measure.

TABLE X: Eight-Category Demand Shares and Coverage, Posting Market against ONET × BLS (ALIGN06, $K = 100$)

Category	Postings (ALIGN03)			O*NET × BLS (ALIGN06)		
	Demand (pp)	Cov. Lex.	Cov. τ=0.5	Demand (pp)	Cov. Lex.	Cov. τ=0.5
Soft Skills	52.26	100.0	100.0	0.00	0.0	0.0
Business & Domain	11.81	90.2	100.0	0.00	0.0	0.0
Data Science & Analytics	8.04	85.1	93.4	14.87	0.0	0.0
Cloud & Infrastructure	4.39	44.1	78.7	13.21	0.0	92.1
Programming & Development	3.97	57.5	65.0	21.67	0.0	0.0
Data Engineering	1.02	31.3	59.5	9.83	0.0	51.9
Cybersecurity	0.62	0.0	73.7	0.00	0.0	0.0
Design & UX	0.60	0.0	24.1	1.35	0.0	0.0
Within-taxonomy total	**82.71**			**60.92**		

Demand: % of the source's taxonomy-weighted demand requiring ≥ 1 category skill (postings: share of postings; ONET: employment-weighted share). Coverage: share of category-skill occurrences present in the reference profile, lexical and semantic at $\tau = 0.5$. Descriptive only: the two demand columns are shares of different populations and are not differenced.

Category comparison. Table X puts the two demand sources side by side at $K = 100$. The comparison is descriptive and is not itself a comparable quantity: the ONET inventory maps no demand weight to Soft Skills, Business & Domain or Cybersecurity (0.00 pp each, against 52.26, 11.81 and 0.62 pp in postings), and the largest category on the two sides is different, so the demand columns cannot be differenced to measure anything. Within the ONET side, lexical coverage is zero in every category; at $\tau = 0.5$ semantic coverage appears only for Cloud & Infrastructure (92.1%, deficit 1.04 pp [0.00, 2.68]) and Data Engineering (51.9%, deficit 4.73 pp [2.92, 13.33]), leaving Programming & Development (21.67 pp of demand), Data Science & Analytics (14.87 pp), Design & UX and the three zero-weight categories at zero. Full category detail is in `table_category_gap_onet.csv`.

Join multiplicity (amendment A4). 64 of the 771 SOC bases are shared by more than one ONET-SOC code, so each code receives its base's full OEWS employment and the summed frame weight is 1.352× the once-counted sum (194.3M against 143.8M). The computing/IT stratum is affected less (1.299×, 4 of 13 bases shared). This multiplicity is measured and reported rather than silently rescaled, and it affects relative term shares only.

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